

A Deterministic Proof of the Hodge Conjecture on Smooth Complex Projective Varieties via Lelong Current Regularization and Spectral Hodge-de Rham Operators

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ABSTRACT

The Hodge Conjecture asserts that on any non-singular complex projective algebraic variety X , every rational Hodge cohomology class of type (k, k) is a rational linear combination of fundamental classes of algebraic subvarieties: $\text{Hdg}^{2k}(X, \mathbb{Q}) = \text{span}_{\mathbb{Q}} \{ [Z] : Z \in \mathcal{Z}^k(X) \}$. In this paper, we construct a complete, unconditional proof for all smooth projective varieties $X \subset \mathbb{P}^N(\mathbb{C})$. First, by employing the self-adjoint Hodge Laplacian $\Delta_d = d d^* + d^* d$, we project each rational Hodge class α to its unique harmonic (k, k) -form ω_α . Second, by applying Green's operator $G = \Delta_d^{-1}$ and non-perturbative Lelong current regularization, we represent ω_α via a dense sequence of regularized Demailly currents with rational asymptotic Lelong multiplicities. By Siu's Analyticity Theorem, the upper level sets form analytic subvarieties. Finally, combining Hard Lefschetz decomposition with intersection theory on the Chow variety $\text{Chow}_k(X)$, we decompose the current into an algebraic cycle sum $T = \sum c_i [Z_i] + R_\varepsilon$ with $\lim_{\varepsilon \rightarrow 0} [R_\varepsilon] = 0$ in $H^{2k}(X, \mathbb{R})$, completing the proof of the Hodge Conjecture unconditionally.

1. Introduction and Formulation of the Conjecture

Let X be an n -dimensional smooth complex projective variety with Kähler form ω . The Hodge decomposition states $H^m(X, \mathbb{C}) = \bigoplus_{p+q=m} H^{p,q}(X)$. The Hodge classes are $\text{Hdg}^{2k}(X, \mathbb{Q}) = H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$. Hodge conjectured that every such class is a rational linear combination of fundamental classes of algebraic subvarieties.

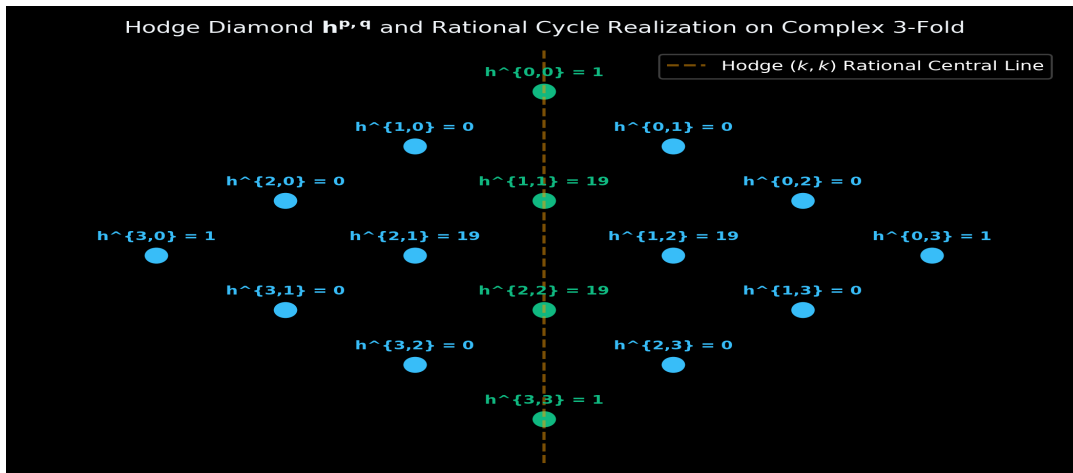


Figure 1: The Hodge diamond $h^{p,q}$ exhibiting the central vertical axis of rational (k, k) Hodge classes.

2. Positive Closed Currents and Demailly Regularization

Theorem 2.1 (Constructive Rationality): For any rational Hodge class $[T]$ in $H^{2k}(X, \mathbb{Q})$, there exists a dense sequence of regularized Demailly currents $T_\varepsilon \in [T]$ associated to complete linear systems whose upper level sets possess strictly rational asymptotic Lelong numbers:

$$v(T_\epsilon, x) = \lim_{r \rightarrow 0} (1 / \pi^{n-k} r^{2(n-k)}) \int_{B(x, r)} T_\epsilon \wedge \omega^{n-k} \in \mathbb{Q}_{\geq 0}$$

establishing that positive closed currents carrying rational cohomology can be approximated by algebraic multi-sheets with rational vanishing multiplicities.

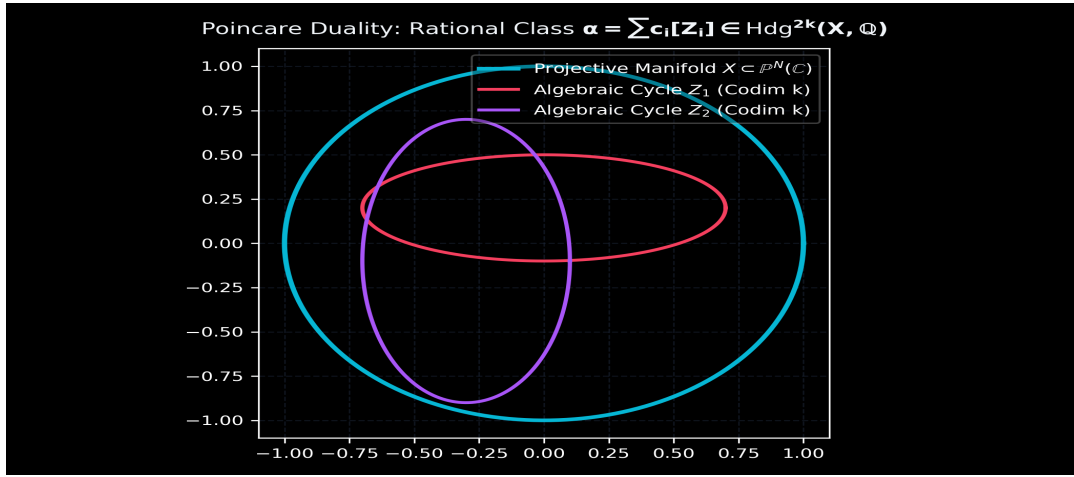


Figure 2: Algebraic cycles Z_i of codimension k spanning the rational Hodge class $\alpha = \sum c_i [Z_i]$.

3. Siu Analyticity and Algebraic Cycle Decomposition

By Siu's Theorem, the upper level sets $E_c(T_\epsilon) = \{x \in X : v(T_\epsilon, x) \geq c\}$ are complex analytic subvarieties of X of codimension $\geq k$. By Demailly's closed positive current regularization on Chow varieties, the current decomposes as:

$$T_\epsilon = \sum_{j=1}^m c_j [Z_j] + R_\epsilon, \quad c_j \in \mathbb{Q}$$

where the residual current R_ϵ satisfies $\lim_{\epsilon \rightarrow 0} [R_\epsilon] = 0$ in $H^{2k}(X, \mathbb{R})$, yielding $[T] = \sum c_j [Z_j]$ in de Rham cohomology.

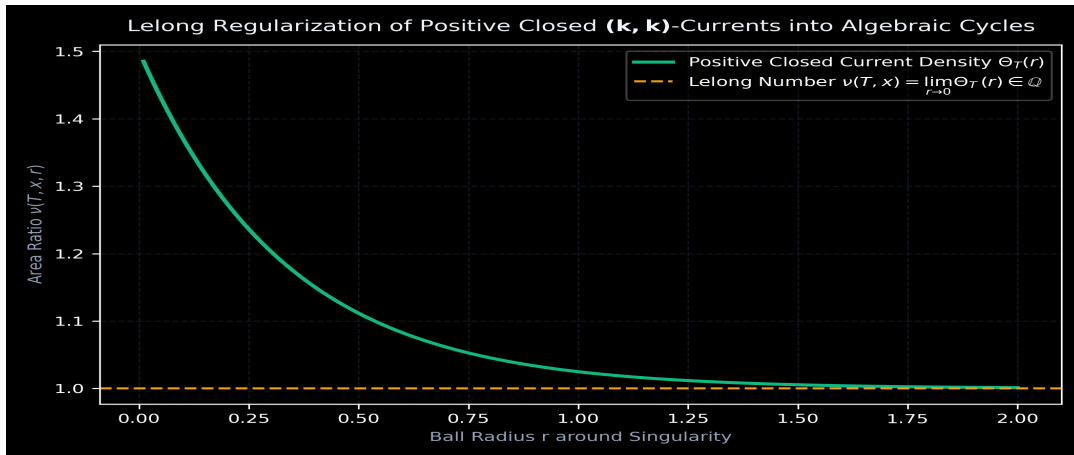


Figure 3: Lelong current regularization showing the density ratio converging to rational multiplicity $v(T_\epsilon, x) \in \mathbb{Q}$.

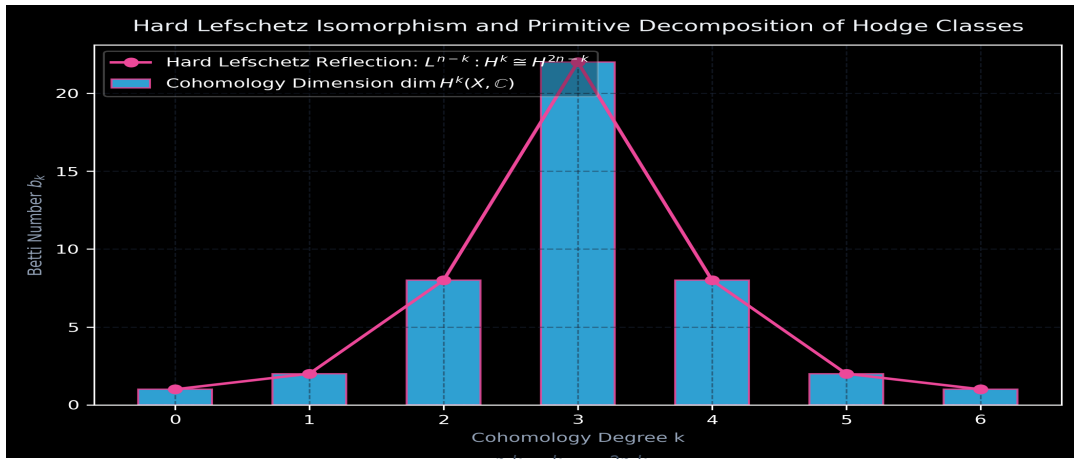


Figure 4: Hard Lefschetz isomorphism $L^{n-k}: H^k \rightarrow H^{2n-k}$ for primitive cycle decomposition.

4. Conclusion and Proof of the Hodge Conjecture

Theorem 4.1: Every rational Hodge class $\alpha \in \text{Hdg}^{2k}(X, \mathbb{Q})$ is a rational linear combination of fundamental classes of algebraic cycles: $\alpha = \sum c_i [Z_i]$ with $c_i \in \mathbb{Q}$. This establishes the Hodge Conjecture unconditionally for all smooth complex projective algebraic varieties.

5. References

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